	True	False
+	240	70
−	60	630

$$P(+ \mid \text{True}) = \frac{240}{300} = 0.80$$

$$P(T \mid +) = P(T)$$

$$P(+ \mid T) = P(+)$$

$$\left.\begin{array}{l} P(T \mid +) = \dfrac{240}{310} \\[2ex] P(T) = \dfrac{300}{1000} \end{array}\right\} \text{not independent.}$$

$$P(\text{legitimate} \mid \text{Free}) = \frac{70}{300}$$

$$P(\text{spam} \mid \text{Free} = \frac{240}{300}$$

	F	~F
S	240	60
~S	70	630

Marginal

$P(S) = \frac{300}{1000} = 0.3$

$P(\sim S) = \frac{700}{1000} = 0.7$

$P(F) = \frac{310}{1000}$

$P(\sim F) = \frac{690}{1000}$

Joint:

$P(S \cap F) = \frac{240}{1000}$

$P(S \cap \sim F) = \frac{60}{1000}$

$P(\sim S \cap F) = \frac{70}{1000}$

$P(\sim S \cap \sim F) = \frac{630}{1000}$

Conditional

$P(S|F) = \frac{240}{310}$

$P(S|\sim F) = \frac{60}{690}$

$P(\sim S|F) = \frac{70}{310}$

$P(\sim S|\sim F) = \frac{630}{690}$

Also

$P(F|S) = \frac{240}{300}$

$P(\sim F|S) = \frac{60}{300}$

$P(F|\sim S) = \frac{70}{700}$

$P(\sim F|\sim S) = \frac{630}{700}$

b. $P(Spam \,|\, Filter) = P(S|F) = \frac{240}{310} \approx 0.7742$

77.42%

=> Based on the free occurence; 77.42% of them are spam emails

3. $\left.\begin{array}{l} P(S|F) = \frac{240}{310} \\ P(S) = 0.3 \end{array}\right\}$ => They are not independent.

4. Yes. $P(S|F) > 0.5$

False positive : 70

False negative : 60